

Juan José Aguado

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PROFILE

AI Engineer with full-stack experience building and shipping end-to-end AI applications in production. Skilled at integrating LLMs, RAG pipelines, and machine learning models into scalable systems on AWS/Azure, with a strong foundation in deep learning, computer vision, and data-driven analytics. Passionate about turning research and unstructured data into reliable, production-grade AI products through experimentation, robust engineering, and automation.

EDUCATION

Pontificia Universidad Javeriana

B.S. in Systems and Computer Engineering (Data Engineering Focus)

Cali, Colombia

Aug. 2020 – Jun. 2025

Universidad Icesi

Master's in Applied Artificial Intelligence

Cali, Colombia

Jul. 2025 – Dec. 2026 (Expected)

EXPERIENCE

Artificial Intelligence Engineer

Cadre AI

Mar. 2026 – Present

San Diego, California, United States (Remote)

- Built and shipped full-stack, end-to-end **AI applications** in production environments.
- Designed and deployed scalable **AWS** infrastructure to support AI workloads and real-time applications.
- Developed custom **telemetry systems** and dashboards to track model performance, system health, and usage.
- Engineered and optimized **prompts** for LLMs to improve output quality and align with business objectives.
- Evaluated and fine-tuned models using structured testing and **validation methodologies**.
- Built and managed **data pipelines** for data collection, preprocessing, and deployment workflows.
- Collaborated cross-functionally to integrate **AI solutions** into production systems.
- Automated repetitive business processes using AI, improving efficiency and reducing manual work.
- Monitored and iterated on deployed systems to ensure scalability, reliability, and continuous improvement.

Data Scientist

Atinna

Oct. 2024 – Feb. 2026

Cali, Colombia

- Designed and automated end-to-end social listening pipelines for multilingual data ingestion, LLM model execution, and analytics reporting, improving productivity by **50%**.
- Developed semantic clustering systems using **embeddings** and **unsupervised learning** to identify conversation themes across millions of social media posts.
- Enhanced classification accuracy for **stance and sentiment analysis** through **few-shot learning** and prompt engineering with **DSPy** and **LangChain**.
- Integrated **Langfuse** to monitor and version prompts in production, ensuring traceability and continuous LLM improvement.
- Built scalable backend services in **Python** connected to **Azure SQL**, supporting dynamic **Power BI** dashboards for real-time client insights.
- Optimized collaborative workflows through **Git**-based versioning and reproducible model deployment pipelines, accelerating iteration cycles.
- Architected and deployed end-to-end LLM workflows for web video extraction, AI description inference, and metadata integration into structured analytics.

Research Assistant – Applied Software Engineering Group

Pontificia Universidad Javeriana

Jun. 2024 – Dec. 2024

Cali, Colombia

- Developed and maintained the university's automated email service using **Java** and **Maven**, enhancing scalability and reliability.
- Validated and tested API endpoints using **Postman**, increasing delivery efficiency and system uptime.
- Collaborated on refactoring efforts to improve code modularity and maintainability across microservices.

Database Administrator

Fundación Casa de Colombia

Jan. 2024 – Jun. 2024

Cali, Colombia

- Led the design and deployment of a PostgreSQL database system to centralize educational and healthcare program data.
- Coordinated software development using **Jira** and **Bitbucket**, ensuring agile progress tracking and version control.

AI Model Evaluator – Generative AI Models

Jun. 2024 – Present

Outlier (Tasker Platform)

Remote

- Assessed the performance of multimodal **generative AI models** (text, audio, and image) for instruction-following, safety, and factual accuracy.
- Delivered detailed evaluations in English, enhancing the reliability and compliance of next-generation AI systems.

PROJECTS

Thesis: Prenatal Cranial Plane Classification using CNNs | *PyTorch*

Jun. – Dec. 2024

- Developed a deep learning model to classify fetal brain MRI images into **transventricular**, **transcerebellum**, and **transthalamic** planes.
- Compared CNN architectures (**ResNet-18**, **DenseNet**, **YOLOv8**) for performance, achieving over **92% accuracy**.
- Preprocessed MRI data with normalization and augmentation to enhance model generalization.
- Validated results against expert medical classifications, demonstrating AI's potential for clinical support.

Email Intelligence System: LLM Agent Classification with RAG | *LangGraph, FastAPI, Docker*

2025

- Designed and implemented a microservices system for intelligent email classification using **LLM agents** orchestrated with **LangGraph**, featuring confidence-based conditional routing across workflow nodes.
- Implemented **RAG (Retrieval-Augmented Generation)** with **ChromaDB** and **sentence-transformers** to retrieve semantically similar examples and enhance low-confidence classifications via few-shot learning.
- Built a **3-microservice architecture** (API Gateway, Classifier, Vector DB) orchestrated with **Docker Compose**, with isolated Docker networks and internal HTTP communication between services.
- Developed **async** endpoints using **FastAPI** with dependency injection, Pydantic type hints, and structured error handling following production best practices.
- Designed a **provider-agnostic LLM system** supporting Gemini, OpenAI, and Anthropic through a factory pattern configurable via environment variables — no code changes required to switch providers.

Smart Farming IoT Platform on AWS | *Python, AWS IoT Core, Lambda, EC2, PostgreSQL*

2024

- Designed an end-to-end IoT platform simulating soil humidity, NPK, and weather sensors, publishing telemetry to **AWS IoT Core** via **MQTT** with hierarchical topic design following IoT best practices.
- Built an event-driven ingestion pipeline with **AWS Lambda** and **SNS** for automated threshold alerts, and a **serverless REST API** using AWS Chalice and API Gateway for sensor management.
- Persisted telemetry data in **PostgreSQL** on EC2, applying event-driven architecture, certificate-based security with IAM, and decoupled pub/sub messaging patterns.

CERTIFICATIONS

AWS Academy Graduate – AWS Academy Cloud Foundations

Cloud Certified Architect Foundations

Cloud Certified Architect Foundations – Early Adopter

Scrum Fundamentals Certified

COURSES

HarvardX CS50 | *Django, HTML, Git, CSS, JavaScript, SQL*

Apr. 2024

- Developed a full-stack web application using **Django**, implementing RESTful APIs and SQL data management.
- Applied secure development practices, testing, and version control with **Git/GitHub**.

SKILLS

Programming: Python, C/C++, SQL, JavaScript, R, Java

Machine Learning / AI: PyTorch, TensorFlow, Scikit-Learn, YOLOv8, ResNet, DenseNet, DSPy, LangChain, LangGraph, RAG, OpenAI API

Data Engineering: Azure SQL, PostgreSQL, Power BI, Docker, Git, FastAPI

Developer Tools: VS Code, Jira, Bitbucket, Langfuse, VirtualBox, Google Colab, Eclipse, Maven

Cloud Platforms: AWS, Azure

Languages: Spanish (Native), English (Advanced)